

东南大学考试卷（期中测试）

课程名称 计算机组织与结构（英文） 考试学期 21-22-2 得分
适用专业 信息科学与工程学院 考试形式 闭卷 考试时间长度 90 分钟

题目	一 (20)	二 (15)	三 (20)	四 (45)	总分
得分					
批阅人					

一、Fill blanks (Total 20 points, 2 points/blank)

- For (random) access, each addressable location in memory has a unique, physically wired-in addressing mechanism. The time to access a given location is independent of the sequence of prior accesses and is constant. (Associative) access is a random access type of memory that enables one to make a comparison of desired bit locations within a word for a specified match, and to do this for all words simultaneously.
- The typical elements of a machine instruction are (operation code), source and destination operand reference, and (next instruction reference).
- Convert the formula $(A + B + (C - D) / E)$ to reverse Polish (ABCD-E/++), Convert the formula $A B C D E + \times \times /$ to infix formula ($A / (B * C * (D + E))$).
- In the three mapping methods of caches, the set associative mapping actually includes the other two techniques. For a cache of 256 lines, if the set size is (256) line/lines it is associative mapping, and if the set size is (1) line/lines it is direct mapping.
- Many computer manufacturers offer a family of computer models, all with the same (architecture) but with differences in (organization). Consequently, the different models in the family have different price and performance characteristics.

二、Translate into Chinese (Total 15 points)

- (7 Points) A wide variety of addressing modes is used in various instruction sets. These include direct, indirect, register, register indirect, and various forms of displacement, in which a register value is added to an address value to produce the operand address.

M.B.R. Memory Buffer Register: Contains a word of data to be written to memory or the word recently read.

- (8 Points) The way out of this dilemma (难题, 困境) is not to rely on a single memory component or technology, but to employ a memory hierarchy. Thus, smaller, more expensive, faster memories are supplemented (补充) by larger, cheaper, slower memories. The key to the success of this organization is decreasing frequency of access of the ^{slower} memory by the processor.

二、Briefly answer questions (Total 20 points)

- (6 Points) Why is a two-stage instruction pipeline unlikely to cut the instruction cycle time in half, compared with the use of no pipeline?

The execution time will generally be longer than the fetch time.

- (6 Points) What is the advantage of PC relative addressing?

Saves address bits in the instruction.

- (8 Points) Four registers are essential to instruction execution: PC, IR, MAR, and MBR. What are the functions of the registers respectively?

PC, program Counter: Contains the address of an instruction to be fetched.

IR, Instruction Register: Contains the instruction most recently fetched.

MAR, Memory address Register: Contains the address of a location in memory

三、Problems (Total 45 points)

1. A direct mapping cache has lines of 16 bytes and a total size of 4k bytes. The main memory of 16-Mbyte is addressed in byte.

- (6 Points) Show the format of main memory addresses (give the bit numbers of Tag, Line, and Word).
- (9 Points) CPU is to read a data with address A531BD (in hex) from the Cache-Memory structure. Describe the reading process (step 1, 2, 3, ...) considering the cases of Hit and Miss.

a) Tag Line Word
12 8 4

b) Tag Line word
A53 1B D

Step1: 1B → Cache

Step2: if tag of cache = A53, Hit

D → cache, read

Step3: if tag of cache ≠ A53, Miss

A531B → memory → 1B line of Cache

D → cache, read

2. (4 Points) Convert the following formula from infix to reverse Polish:

$$X = (A \times (B + C)) - D / (E - F)$$

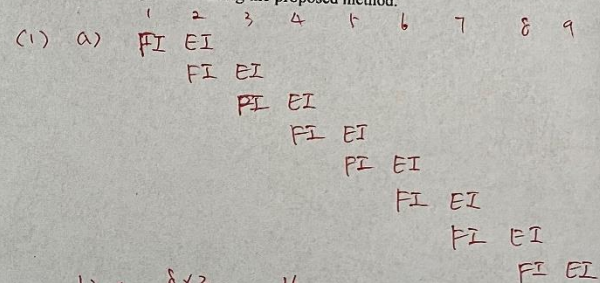
(6 Points) Write programs to compute the above formula using stack.

ABC+*DEF-/-

PUSH A	PUSH D
PUSH B	PUSH E
PUSH C	PUSH F
ADD	SUB
MUL	DIV
	SUB
	POP X

3. To perform a sequence of 8 instructions (without branch taken) using pipeline, each instruction is divided into 2 stages, **Fetch**, and **Execute**,

- (6 Points) Assume that both of the two stages require 1 time unit.
 - Draw a diagram to illustrate the pipeline
 - Analyze the speed up factor of the pipeline
- (6 Points) Assume that the two stages require 1 and 2 time units respectively.
 - Draw a diagram to illustrate the pipeline.
 - Analyze the speed up factor of the pipeline.
 - Propose a method to improve the performance of the pipeline, and calculate the speed up factors using the proposed method.



$$b) S = \frac{8 \times 2}{9} = \frac{16}{9}$$

4. (8 Points) The following code is part of a simulation program, it contains a loop L1. Assume that we use one historical information bit to predict the branch dynamically. How many times of branch prediction errors will occur in the execution of this program?

Assume that the initial taken/not taken switch is set to be "branch not taken". ("branch taken" means to return to the beginning of the loop)

	N	predict	Execution	Error
N = 0;				
M = 0;				
L1: for N = 0:1000	0	NT	T	1
XXXXXXX;	1	T	T	0
XXXXXXX;				
XXXXXXX;				
end	...			
XXXXX;	1000	T	NT	1

Two branch prediction errors.

